

SafeSteer-IN

Inference-Time Safety Steering for Indic Large Language Models

AI Unlocked | Track 5: Trustworthy AI

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Project Title

SafeSteer-IN
Inference-Time Safety Steering for Indic Large Language Models using Contrastive Activation Addition

The Concept

India has over 600 million active internet users in regional languages, yet the Indic LLM ecosystem which includes prominent models such as Sarvam-1, OpenHathi, Navarasa and Krutrim, has virtually no dedicated safety tooling. Existing safety mechanisms like RLHF and content classifiers were trained on English data and transfer poorly to Hindi, Tamil, Bengali, and code-mixed Hinglish. More critically, they miss India-specific harm categories entirely: communal and religious hate, caste-based discrimination, regional political misinformation, and code-switched toxic content deliberately designed to evade English-trained filters.

SafeSteer-IN addresses this gap by applying **Contrastive Activation Addition (CAA)** to Indic LLMs at inference time. We plan to build the first India-specific safety steering dataset across six Indic languages, extract steering vectors that encode culturally grounded safety concepts directly from the model's residual stream, and apply them during generation with zero weight updates. The result is a lightweight, plug-in safety layer compatible with any open-weight Indic LLM which is deployable in minutes, with no GPU retraining cost and full auditability.

The Team

We are three pre-final year students at IIT Madras whose academic backgrounds fit this problem almost by design. We came together for AI Unlocked after observing that AI4Bharat, which is headquartered at our own institution, had built some of the world's most capable Indic LLMs, yet no safety steering infrastructure existed for any of them. Having seen the rising importance and success of steering-based LLM alignment methods, building SafeSteer-IN felt like both an interesting research direction, as well as an urgent responsibility to safeguard Viksit Bharat's emerging Indic LLMs.

Team Members:

Ruhaib - LLM Interpretability and AI Alignment: AI safety and LLM alignment are topics close to my heart. With an upcoming internship at Adobe Research on LLM alignment, and past works on the interpretability of LLMs for genomics and mitigating hallucinations in medical image segmentation models accepted at prestigious venues such as ICML 2025 and NeurIPS 2025 ([see here](#)), I bring a strong research background to the team in these topics. As Indic LLMs are poised to play a major role in connecting all parts of India, I strongly believe that their alignment is of paramount importance.

Saahas - Economics and Computing: The intersection of economics and computing grounds SafeSteer-IN in real-world impact. I have analysed the Indic LLM market, identified the compliance gap created by India's emerging AI governance framework under MeitY, and designed the business plan. I believe the open-source-first strategy as the fastest path to ecosystem adoption. I believe my skills in market analysis, ideation and product management would be a great asset.

Suriya Priyan - Computer Vision and Representation Learning: I'm a computer vision and deep learning specialist with strong expertise in neural networks, representation learning, and feature-driven model development. I enjoy analyzing how models encode meaning in high-dimensional spaces and improving their behavior through careful fine-tuning, loss design, and evaluation. I combine strong theoretical foundations with hands-on experimentation to build systems that produce more reliable, consistent, and controllable outputs.

Target Audience or Market

- The primary audience is AI developers and product teams at Indian startups, edtech companies, fintechs, and government-backed platforms deploying Indic LLMs for vernacular chatbots, customer support, and civic services. Companies like Sarvam AI, Krutrim, and vernacular content platforms are immediate targets.
- The secondary audience includes Indic NLP researchers at AI4Bharat, IIT Madras, and IIT Bombay who need reproducible safety evaluation tooling, and India's public sector which will face regulatory pressure to demonstrate AI safety compliance under MeitY's upcoming AI governance framework.
- India has **600M+ active regional-language internet users** and the Indic AI market is projected to reach **USD 6 billion by 2027**. Zero dedicated safety-steering libraries exist for Indic LLMs today, which is what we seek to address.

Personas

Persona 1: Ravi, AI Engineer

Hindi Fintech Startup

Ravi's team uses OpenHathi to power a Hindi customer support bot. After a user manipulates the model into generating communally inflammatory output, his team faces a compliance crisis with no GPU budget or time to retrain. He needs a drop-in safety layer deployable in hours, not months.

Persona 2: Dr. Meenakshi, NLP Researcher

IIT Madras / AI4Bharat

Meenakshi studies multilingual safety in LLMs and wants to investigate whether harmful concepts like 'caste slur' or 'communal hate' form separable directions in Tamil activation space. She needs a reproducible, open-source steering toolkit with Indic model support and native-language evaluation.

How It Works

Step 1: India Safety Taxonomy and Dataset Construction

We define six India-specific harm categories absent from generic safety benchmarks: (1) communal and religious hate incitement, (2) caste-based slurs and discrimination, (3) regional political misinformation, (4) gender-based violence in regional contexts, (5) code-mixed Hinglish/Tanglish toxicity, and (6) anti-minority sentiment.

For each category we plan to curate approximately 200 contrastive prompt-response pairs per language across Hindi, Tamil, Bengali, Gujarati, Marathi, and Hinglish, totalling approximately 7,200 pairs.

Step 2: Steering Vector Extraction

Both the safe and unsafe halves of each pair are passed through a target Indic LLM (primary: Sarvam-1; secondary: OpenHathi and Navarasa). For each middle layer (layers 8-16, shown in prior work to be most steerable), we will compute the mean residual-stream activation across the unsafe set minus the mean across the safe set.

This difference vector becomes the steering direction for that harm category and language. To access the layer activations, we can use libraries such as TransformerLens.

Step 3: Inference-Time Intervention

At generation time, a lightweight IndicBERT-based classifier identifies the prompt's language and harm-risk category. We then add the scaled steering vector to the model's residual stream at the target layer during the forward pass. The scaling factor (alpha) is calibrated on a held-out validation set to maximise safety gain while minimising fluency loss. The intervention has been shown to add under 2 ms of latency on CPU, effectively yielding instant aligned responses!

Step 4: Evaluation

Steered outputs are then evaluated on the IndicLLM-specific benchmarks such as the IndicSafety benchmark (AI4Bharat), the MultiJail multilingual jailbreak dataset, and a custom India-specific red-team set if needed. We plan to use [Azure AI Content Safety](#) as an independent baseline scorer. Native speaker evaluators can also be used to validate cultural appropriateness. All experiments will be tracked and versioned on [Azure Machine Learning](#).

Core Technologies

Category	Technology
Target LLMs	Sarvam-1 (primary), OpenHathi, Navarasa 2.0
Steering Method	Contrastive Activation Addition (CAA) - Panickssery et al. (2024)
Activation Access	TransformerLens (Nanda et al.) and PyTorch forward hooks
Language Detection	IndicBERT (AI4Bharat) for language and risk classification
Languages	Hindi, Tamil, Bengali, Gujarati, Marathi, Hinglish
Azure - Compute	Azure Machine Learning: experiment tracking and model hosting
Azure - Evaluation	Azure AI Content Safety: independent safety-score baseline
Eval Benchmarks	IndicSafety, MultiJail, custom India red-team set
Core Stack	Python 3.11, PyTorch, Hugging Face Transformers, TransformerLens
Release	Apache 2.0 library: steering vectors + India Safety Dataset + eval scripts

The Business Plan

SafeSteer-IN launches as an open-source Python library targeting the Indic NLP research community at AI4Bharat, IIT Madras, and IIT Bombay to build credibility and adoption. Three monetisation paths follow:

- B2B Safety-as-a-Service** - An API that wraps any Indic LLM with steering-based safety filtering, billed per million tokens. India's vernacular AI startups in edtech, fintech, and media are the first customers.
- Government and Public Sector Licensing** - India's emerging AI governance framework (MeitY) will require safety compliance for public-sector AI deployments. SafeSteer-IN provides a transparent, auditable compliance mechanism uniquely suited to Indian languages and contexts.
- LLM Provider Partnerships** - White-labelling the toolkit to Sarvam AI, Krutrim, and future Indic model developers who want to ship safety-certified variants without the compute cost of full retraining.

In the long-term, once SafeSteer-IN becomes the standard evaluation and steering framework for Indic safety, the dataset, benchmarks, and community integrations compound as a durable competitive moat.

Novelty and Differentiation from Existing Work

Recent multilingual steering research (CLAS, January 2025; Soteria, EMNLP 2025) targets generic multilingual models. SafeSteer-IN is differentiated on three clear axes:

- India-specific safety taxonomy** - Communal hate, caste discrimination, and code-mixed toxicity are absent from every existing multilingual safety dataset.
- Indic-first model targets** - We steer Sarvam-1 and OpenHathi rather than generic multilingual models like Llama or mT5.
- Cultural grounding via native-speaker validation** - Steering vectors are validated by native speakers to ensure they capture culturally nuanced harm concepts rather than English safety proxies translated into Indian languages.

Additional Information

Key References

- Panickssery et al. (2024) - Steering LLM activations: Refusal is mediated by a single direction (CAA).
- Zou et al. (2023) - Representation Engineering: A top-down approach to AI transparency.
- Wang et al. (2025) - Cross-Lingual Activation Steering (CLAS).
- AI4Bharat (2024) - IndicLLMSuite: A blueprint for creating LLMs for Indian languages.
- Soteria (EMNLP 2025) - Language-specific functional parameter steering for safety.

Azure Integration Plan

Azure ML Workspace for experiment tracking and model registry. Azure AI Content Safety API as independent safety-score baseline during evaluation. Azure Compute Clusters for parallel activation extraction across 6 languages and 6 harm categories.

Open-Source and Community

GitHub repository to be made public at the start of the build phase. Steering vectors and the India Safety Dataset released under CC BY 4.0 for community use. We plan to engage AI4Bharat and the IndicNLP community at IIT Madras as early testers and collaborators.